CTMSDFIS SII P4: Artificial Intelligence

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		Total
					Internal Exams				University Exams		University Exams (LPW)		
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
03	00	00	03	03	25	00:45	50	01:30	100	03:00	-	-	200

* Note: TA-2will be in form of assignments or workshops.

Objectives

1. To understand concept of Artificial Intelligence, Machine Learning and Deep Learning
2. To learn various Machine Learning and Deep Learning techniques
3. To create models to understand applications of AI
4. To learn role of ML/DL in Information Security

UNIT-I
Introduction to Mathematics for ML and Python

Python: Setting up Environment, Basic Python Commands, Creating Python Scripts, Conditions, Loops, List, Dictionary, User Defined Functions, Introduction to Anaconda, Working with NumPy, Pandas and Matplotlib. Mathematics for ML: Vectors, Matrices, Linear Equations, Mean, Median, Mod, Standard Deviation and Variance, Probability, Correlation, Regression, Handling and Representing Data.

UNIT-II
Machine Learning (ML)

Definition and History of AI, Defining Machine Learning, Applications of ML, Issues and Challenges in ML, Types of ML. Basics of Supervised Learning, Prediction, Classification, Understanding Datasets, Feature Selection, Feature Normalization, Data Cleaning, Training, Testing & Validation Sets, Different Models of Supervised Learning, Hyperparameters, Measuring Performance, Accuracy and Loss Underfitting & Overfitting, Basics of Unsupervised Learning, Different Models of Unsupervised Learning.

UNIT-III
Neural Network

Understanding Biological Brain, Defining Artificial Neural Network (ANN), Applications of ANN & DL. Defining & Building a Perceptron, Feed Forward, Back propagation, Single-layer & Multi-layer ANNs, building an ANN Model, Activation & Loss Functions, Compiling & Evaluating a Model. Convolutional Neural Networks

(CNN): Understanding Convolutions, Pooling, Building & Fitting CNN Models, Evaluating Model Performance. Recurrent Neural Networks (RNN): Basic RNN Architecture, Applications of RNN, Building & Fitting RNN Models, Evaluating Model Performance. Long Short-Term Memory Networks (LSTM): LSTM Network Architecture, Understanding LSTM, Building LSTMs

UNIT-IV
Computer Vision and Natural Language Processing

Computer Vision: Introduction, Object Detection and Image Segmentation, Detecting and Recognizing Faces, Tracking Objects, Pattern Recognition. Natural Language Processing (NLP): Introduction, Language as Data, Building Custom Corpus, Text Vectorization & Transformation, Classification for Text Analysis, Clustering for text Similarity, Context-Aware Text Analysis, Text Visualization.

UNIT-V
ML/DL for Cyber Security

Introduction to Role of ML in Cyber Security, Malware Detection & Classification, Anomaly Detection, Pen Testing using ML, Social Engineering, ML based Intrusion Detection and other Applications of ML in Cyber Security.

Reference Books

1. Mathematics for Machine Learning 1st Edition by Marc Peter Deisenroth
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems 2nd Edition by AurélienGéron
3. Python Machine Learning: Machine Learning and Deep Learning with Python, scikit- learn, and TensorFlow 2, 3rd Edition by Sebastian Raschka and Vahid Mirjalili
4. Hands-On Neural Networks with Keras: Design and create neural networks using deep learning and artificial intelligence principles 1st Editionby NiloyPurkait
5. Deep Learning with Keras: Implementing deep learning models and neural networks with the power of Python by Antonio Gulli, Sujit Pal
6. Practical Machine Learning for Computer Vision 1st Edition by Valliappa Lakshmanan, Martin Görner and Ryan Gillard
7. Learning OpenCV 4 Computer Vision with Python 3: Get to grips with tools, techniques, and algorithms for computer vision and machine learning, 3rd Edition by Joseph Howseand Joe Minichino
8. Natural Language Processing in Action: Understanding, analyzing, and generating text with Python 1st Editionby Hobson Lane, Hannes Hapke and Cole Howard.
9. Machine Learning for Cybersecurity Cookbook: Over 80 recipes on how to

implement machine learning algorithms for building security systems using Pythonby Emmanuel Tsukerman.

10. Hands-On Machine Learning for Cybersecurity: Safeguard your system by making your machines intelligent using the Python ecosystem by Soma Halder (Author), Sinan Özdemir